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Intra-acting bio-object: A posthuman approach to the player–game relation

ABSTRACT

The main aim of this article is to explore how posthuman values and premises can change the approach to video game research, in terms of reframing the relation between game and player as a meaning-making process. The idea of the bio-object, which originated in Tadeusz Kantor's avant-garde theatre, is introduced and reread in the context of the critical posthumanism and new materialism of Karen Barad, especially her concept of intra-action. By meshing together Kantor's and Barad's ideas, a framework is developed for conceptualizing the bond between the player and the video game object, pointing out how their constant rivalry is not only resolved in meaning-generative tension, but also intra-actively shapes their ontic borders. The game and the player become equal in this new unity, and the video game object stops being perceived as a secondary to the player and can be analysed as the equal partner in this relation.

KEYWORDS

bio-object
intra-action
posthumanism
Tadeusz Kantor
Karen Barad
player–game
relationship

INTRODUCTION

Posthumanist approaches are gaining more and more attention in the field of game studies. As 'posthumanism' refers to a broad category, the research itself also contains a wide spread of topics, from non-human play (Wirman 2014; Westerlaken and Gualeni 2016) and research methods based on Latourian actor-network theory (Giddings 2008; Jessen and Jessen 2014), to the ontology of the game object (Bogost 2012; Fizek 2018). However, there is still a need for work that focuses on the creation of meaning inside the player–game relation and the play process itself, which, at the same time, would emphasize the ethical foundation of a posthuman approach focused on the human relation with technology. With that in mind, I will follow the premises of the critical posthumanism of Karen Barad (2007), especially her concept of intra-actions. However, to focus more on the aesthetical side of the relation and the production of meanings within it, I shall introduce the idea of the bio-object proposed by Tadeusz Kantor (2004). By combining both of those approaches, one from the field of philosophy, the other from theatre studies, together, I will be able to theorize the player–game relation in a way that simultaneously emphasizes both the equality and the uniqueness of human and non-human actants during gameplay, as well as their ability to create meanings together.

WHAT IS POSTHUMANISM?

The way in which I shall use the term 'posthumanism' in this article should be specified. The term can be confusing in the breadth of its application, and in how many different, sometimes very contradictory approaches can be found in this category – from the post-dualistic approach of (philosophical, cultural, critical) posthumanism, through the feminism of new materialism, to human-centric transhumanism (Ferrando 2013: 26). I perceive posthumanism as a more affirmative alternative to traditional humanism and antihumanism in its search for a new way to describe the human subject without focusing on the 'crisis of the human condition' (Braidotti 2014: 100–01). What seems to be the most important premise of this approach is a non-anthropocentric view of reality, in which the human not only stops being the central figure that governs the surrounding environment, but is also deeply connected to other entities – both living and non-living – that influence each other in this entanglement. From this perspective, it is very easy to see that posthumanism is very much not only a methodological statement, but also an ethical one. By shifting the human position from the centre, we start to listen to the voices of non-human actants that are not only part of the same reality, but are also its co-creators (see Latour 2005; Barad 2007; Wolfe 2009; Haraway 2015). However, as Rosi Braidotti (2013) points out in *The Posthuman*, there is still need for a discussion on what a posthuman ethics would really mean, and on how we should apply posthuman philosophy in a way that preserves its core premises. This seems to be a core problem of any form of philosophical or critical posthumanism: can we escape an anthropocentric language of analysis, when we are still humans (Boyd 2015: 20)? Of course, a fully affirmative response would be a rather utopian one, but it is our responsibility as researchers to try to get as close to it as we can.

This question is just as important in the context of video games, which are not only created by humans, but – which is particularly significant in this context – in most cases are created in a way that places the human player in the centre of the experience. Most mainstream games locate the player at the

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heart of the gameworld, which is treated usually as a digital playground where she can express herself within the boundaries of the digital object. This feeling of control is emphasized by the creation of digital environments that highlight human agency and prioritize the main senses used by humans – mainly sight, touch and hearing.

Of course, the limitations, and the potentially illusory nature of the centrality of the player's position, have also been noted. As Daniel Muriel and Gary Crawford point out, digital games are a reflection of the neo-liberal discourse that governs social reality. The player seems to be in a central and controlling position, but, in fact, they are always being controlled by the game's boundaries. As such, this freedom is only apparent – neo-liberal mechanisms allow the player to be successful, but, at the same time, the responsibility for the failure is also the player's (Muriel and Crawford 2020: 140–46). Similar conclusions were also made by Nick Dyer-Witheford and Greig de Peuter (2009), on the basis of a Marxist theoretical approach.

Nonetheless, by and large, the anthropocentric character of game design has tended to be reflected in game studies. Many key theoretical concepts, such as the implied player (Aarseth 2007), the avatar/playable figure (Bayliss 2007; Jørgensen 2009; Klevjer 2012; Vella 2015), immersion (Calleja 2011) or even agency (Murray 1997; Mateas 2004; Wardrip-Fruin et al. 2009), were developed as a means to describe the human player's involvement with the digital worlds. If, as researchers, we decide to stand only in this central position – experiencing the 'game as played' (Leino 2009) – there is the distinct possibility that we would miss an important mechanism or phenomenon, which would make our reflection incomplete. Therefore, it is worth recognizing this problem and trying to shift the perspective from player-centred research towards a relational approach.

While I would still perceive a connection between the player and the video game in the context of a relational network – similar to Thomas Apperley (2010) and Seth Giddings (2008) with Helen Kennedy (Giddings and Kennedy 2008) – I would also want to take a step away from the science and technology studies paradigm and make a shift towards the critical posthumanism and new materialism of Karen Barad. In this way, I will be able to not only focus on how the player and the game object exist inside a complex, multi-actant relational network, but also zoom in closer to the intimacy of single-player gameplay, where two autonomous actants define each other's boundaries and properties on a very material level, staying together yet apart.

This approach is similar to the theory presented by Brendan Keogh (2018) in his book *A Play of Bodies: How We Perceive Videogames*. To emphasize the performative relationship between the player and the game, he combines in his explorations the works of N. Katherine Hayles, Donna Haraway and Maurice Merleau-Ponty. The player and the game become the parts of a complex hybrid that includes both human (the body of the player) and non-human elements (the body of the game, including the hardware, the software, the audio-visual layer and everything that builds the experience of play). The status of the subjectivity of this hybrid creation is important here: the 'player cannot be considered before or distinct from the video game but instead reflexively as producing the video game experience that in turn produces the player' (Keogh 2018: 27). Both the player and the game do not have stable boundaries on their own, but become more defined through their connection. They change and evolve connected together. Keogh uses Hayles' idea of embodied textuality to explain how this human and non-human bodily coupling is

responsible for creating meanings. The human player, by this understanding, does not have absolute agency: they are just a part of the cybernetic circuit in which agency moves between different actants that may or not be granted its power (Keogh 2018: 24).

Despite the fact that the theory formulated by Keogh seems to be a holistic posthuman approach, there are some parts that warrant further exploration – especially the mechanism of the production of meaning and agency, whose meaning and erratic status need more explanation. Therefore, I will not only follow the premises of Barad's new materialistic theory, but also explore the aforementioned issues through the lenses of the aesthetics and theories of the Polish artist and dramatist Kantor. In order to address the specificities of the relation between the player and the game, I would like to build my analysis upon his concept of the bio-object (Kantor 2004: 363–406), not only because, as I shall argue, it aligns with the premises of the critical posthumanist and new materialist approach I have already outlined, but also because, while remaining within this posthumanist paradigm, it more closely reflects the nature of the video game (and the various relations between it and the player) as an object of aesthetic consideration. It can be perceived as a transition from Baradian meaning that emerges directly from the matter to a meaning-generative tension rooted in performative relations between different types of bodies.

GAME/R-PLAY/ER-BIO-OBJECT

The idea of the bio-object emerged from Kantor's aesthetical explorations concerning the nature of art (especially theatre) and its inextricable connection to life and reality. While his style and approach to art evolved organically over time, we still can distinguish the topics he was enduringly interested in. One of these was the idea of objects, their meaning and their place in the surrounding reality. He was interested in how things – ready-made, taken from everyday life, sometimes garbage-like – can become 'L'OBJET D'ART' (Kantor 2004: 397, 415–16), and in how autonomous objects can be perceived in an aesthetic context. As Ewa Domańska pointed out, Kantor's objects are in fact foreign to the human mind, because they do not mimic anything – they escape the process of anthropomorphization or symbolization (Domańska 2008: 19–20). We can find these objects not only in Kantor's artistic installations, but also in his plays. Here, these autonomous, worn-out and grounded-in-reality objects started to be not just simple props, but a part of a structure he termed the bio-object.

Kantor coined the notion of the bio-object to describe the special relation between the actor and the stage object that is established during the performance of the play. The actor uses this object on stage, but not as a prop that she can liberally use as she wishes. Objects, as Kantor put it, 'created an indivisible totality with the actors' (Kantor cited in Kobialka 2009: 359). The actor not only animates the object, but also becoming its 'living organs' (Kantor 2004: 397). This is even visible in the names that can be found in the script of the play, which seem more like a description of the hybrid and its functions than a simple tag: the Man with a Sack and its Unknown Content, the Woman Drowned in the Bathtub, the Helpless Man with a Table. The actor makes the object 'alive', and, without them, it is just a shell on the stage, but at the same time it is the object that defines the moves and motives of the actor, as well as organizes the whole stage materiality (Klossowicz 1991: 31). Like

Barad's intra-actively constituted 'objects', the bio-object is defined, and gains its shape, its boundaries and its life, through this relational encounter.

This situation can also be applied to an analysis of the bond between the player and the video game in the moment of gameplay: the presence of the player animates the game, but it is the game that creates a frame for the player's actions, which also gives this bond a temporal dimension. In order to change and progress, both the player and the game need to work together. This is true for most existing genres of games, with, perhaps, the exception of MMORPGs, always ready for a play session, and idle games, where the presence of the player is mostly unnecessary (Fizek 2018). However, what is special about the idea of the bio-object is that both the human and non-human actant are equal in this qualitative new unity and, as equals, they are both the main conduit of the play's meaning. This is possible because of the unstable nature of this connection: there is a continuous struggle inside the bio-object about which side – human or non-human – will gain advantage over the other; either the actor/player dominates the object and uses it as she wishes, or the object/game imposes itself over the human and confines their movements. In established interpretations of Kantor's theatre, for example in the work of Krzysztof Pleśniarowicz, this never-ending rivalry between two individual forces is perceived as a conflict between mind and matter that builds tensions which, in the end, resolve in the production of new meanings (1990: 34–35). However, this Cartesian division can be reread in a more post-human way, when we realize that Kantor's goal was to equalize both parts of the bio-object. He let non-human elements be a crucial part of the creation of meanings on stage by challenging the actor's perception with the overwhelming presence of the non-human object. The actor could not form a mental bond with the role she needed to portray and control the stage situation this way. She had to face the non-human other not on the level of her dramatic role, but on the level of her material body and its limits.

Examples of this struggle can be observed in stage situations in which the materiality of the object imposes itself over the actor who is trying to manipulate it (e.g. the actor can trip over the ragged body of a mannequin, or an elaborate mechanical structure might just get stuck in the middle of an action). However, this struggle is arguably even more noticeable in the situation of digital game play: when the player is skilful, we can say that they are in a more dominant position, but the moment the game level becomes too hard to beat, the tipping point moves towards the game object. This rivalry becomes most prominent when the player starts to subversively explore the boundaries of the game environment, by, for instance, engaging in transgressive behaviours or forms of counterplay. This could include activities such as making some changes to the game script, by modding the game (Sotamaa 2010), hacking the hardware (Meades 2015), using cheat codes to gain advantages (Consalvo 2007) or even playing against the rules of the diegetic world conveyed by the game object (Pötzsch 2019). These are only a few of the ways in which the player can (re)establish human dominance over the game object.

On the other hand, the game object can also behave subversively, anticipating some of the player's exploitative behaviour. For example, in the games in the *Animal Crossing* series, there is a character named Mr Resetti, who is an anthropomorphized representation of the mechanism that tries to stop the player from resetting the game without saving it. He not only very angrily informs the player about the consequences of such behaviour on the whole system, but also fakes resetting the game in order to scare the player. Another

example from this series is the mechanisms that block the player from using pirated versions of the game. The game informs the player about the seriousness of the issues in the form of a short, humorous cut-scene implying that the player is going to jail, following which the software is effectively turned off.

In such cases, the game regains a dominant position by exposing the presence of its digital materiality. Another situation we could consider in the same light is that of the glitch – often something unplanned and unexpected that can interfere with the player's actions or perceptions of the game environment (e.g. by breaking narrative involvement). The most common examples of this phenomenon are graphical glitches like deformation of character bodies, floating objects or missing textures. However, glitches can also be more complex. In *The Elder Scrolls V: Skyrim* (Bethesda Game Studios 2011), for example, there is the possibility for the player to buy a house. One of the many commodities that the player receives with a property is mannequins that can be used for displaying armour. Sometimes, however, they do not stand still, but either change their pose or, in more extreme cases, start to move around, often following the player-character. This happens because, as in-game objects, they are an NPC model with a mannequin skin that is scripted to stand still – and that script tends to glitch, creating this unintentionally spooky gameplay experience.

When faced with such glitches, the player is pushed to reinterpret what they used to know about the game. Suddenly, the game becomes a very visible technological artefact that seems to have its own autonomy and does not want to be tamed (Janik 2017). The process of play, then, is a performative, ongoing process that reshapes both the player and the game, in which the dominant position of neither actant is fixed.

This is the moment when we should ask the question about the operational rules determining how it is possible that, in this rivalry, new meanings are produced. In Kantor's theory, this happens because of two reasons. Firstly, meaning is produced due to the aforementioned instability of the connection between actants – the bond between actor and object is not exactly stable and the dominant position is not fixed. Second, the production of meaning is possible because of the specific status of the Kantorian object, which is not simply defined by its given, human functionality, but has the capacity to define and transform the human actants it engages. It is possible because of its autonomy. Even if the object, the *emballage*, is deprived of human presence, it is still not just a prop or costume. It can be an empty shell, but, as Mischa Twitchin noticed, it is still a model of a certain idea, which 'represents "itself" beyond any particular performance in which it makes an appearance' (2016: 241).

In various interpretations of Kantor's writing, this process is often perceived as an objectification of the human actor and an attendant subjectification of the object, which is sometimes perceived as degrading for the human condition (see Koch-Butryn 2002; Miklaszewski 2014). However, I believe that the notion of the bio-object needs to be reread in the context of critical posthumanism, to shake off the anthropocentric inclination of the supposed depersonalization of the actor. The extreme attention that Kantor pays to the simplest and most common objects suggests that it is not a case of the human become something lesser, but of things become equally inclined to uncover truths about reality, including such abstract ideas as death, memory and love. Therefore – despite some possible differences between her and Kantor's views on the subject of materiality, which require further discussion – I would like to

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use Barad's (2007) concept of intra-actions to fill in the vague spots in Kantor's writing, especially the one regarding the performative character of the relationship between human and non-human actant, and arrive at an understanding of the idea of the bio-object that can be applied to digital play.

INTRA-ACTIONS AND BIO-OBJECTS

For Karen Barad, reality, and materiality, is not something that is given and stable. It is constituted by phenomena, which she perceives as 'relations without preexisting relata' (Barad 2007: 139). The components of the phenomena (subject/objects/entities) do not have predetermined boundaries or properties. The agential cut that distinguishes and shapes components inside these phenomena is only possible on the basis of a specific agential intra-action. In other words, by this understanding, reality is composed of 'things-in-phenomena' (2007: 140) that gain their individual properties only through specific material-discursive practices, which are 'ongoing agential intra-actions of the world' (2007: 149). For Barad, relations are not suspended in a vacuum, but emerge from the sea of phenomena, which should be perceived as matter that remains in a perpetual state of action. This matter, however, is not something stable and unchanging, but is a process, an existing being, which has the potential for continuous doing/undoing (Barad 2012: 14–16). Moreover, for Barad, this moment before relations happens – a nothingness, a void – is a realm of infinite potentiality, open to constant change: 'the void is a living tension, a desiring orientation towards being/becoming' (2012: 13). Thus, even in nothingness there is a potential for the emergence of dynamic matter, from which phenomena can then form.

In Barad's onto-epistemological agential realism, intra-actions, as the transformative power that produces and shapes the given actants, replaces the concept of interactions, because, by this understanding, there are no determined entities preceding relations that can act between each other. This also changes the status of agency, which is not something that actants have and can use, but rather is a dynamic force that happens between them. By not differentiating between human and non-human agency, Barad wants to escape the anthropocentric tendencies that can unintentionally appear when using the notion of agency in a traditional understanding, while, at the same time, also emphasizing its transformative power (Barad 2007: 149).

This transformative power can manifest itself in various ways within the process of play, shaping the player and the video game object through the relation between them. On the very basic, material level, the state of the game object is transformed when the player makes changes inside the game environment – finding resources, eliminating enemies, solving riddles, and so on. In the opposite direction, the player's hand-eye coordination might improve because of the hours she spends playing a game, which could also result in some muscle pain or even the onset of repetitive strain injury due to the focused physical movement the game requires of them – both of which represent physical transformations wrought upon the human player during the intra-action of play with the game. This can also happen on a more cognitive and physiological level, when the game moves us emotionally by exposing us to new experiences that trigger bodily reactions. A frightening horror game, or a very exciting level in an action game, that triggers a feeling of fear or arousal that is deeply connected with the body of the player – we start to sweat. This physiological occurrence can be measured biometrically, using techniques like

Galvanic Skin Response (GSR). With GSR we can measure how the conductivity of human skin changes with the presence of different emotional stimuli, including the gameplay experience (Schott et al. 2014: 7). The transformation here is not always visible for the player, but nevertheless their body was transformed on a microscopic level.

We are no longer an anonymous model, an implied player (Aarseth 2007), that was born in the head of the designers and marketing specialists during the long process of developing a game. With every action we take inside the game environment, and every action taken upon us by the game, we slowly become us-the-player, a unique entity that came to play. Of course, we do not stop being a human individual, constantly being shaped by multiple different material-discursive practices, from the basic atomic level to the grand cultural narratives. But intra-acting with a game gave us new properties and new boundaries as we are shaped into a player during the moment of play.

The same goes for the video game object. It is transformed into what we understand as a game on various different levels that are sometimes not even visible to us. It could start as an idea in a designer's mind, before slowly becoming a digital object intended to convey an environment, and finally to be installed on our gaming machines. It might be dormant, taking space on our hard drive, existing through technological intra-actions that are often ignored because of their infinitesimal visibility to a common computer user. However, in the moment of the intra-activity of play, the video game is transformed from a digital object/computer application to a playable artefact that has materiality intertwined with the software's behavioural processes (Leino 2012). New properties are manifested and explored in the moment that the game's relationship with player starts. However, the real power of intra-activity leads not only to the transformation of each of the actants, but also to making them distinct. We become the player, with our own conviction and experiences, and the game object becomes different from other copies thanks to its being installed on a different machine and changed in a specific way during the player's explorations. The intra-active power of play shapes both of us into uniqueness.

The player and the game object are constantly reconfiguring each other, but also do not emerge in this specific form outside this connection. As Linus de Petris and Anders Falk rightly point out when interpreting game(play) in Barad's terminology: 'a gamer or a game is not made meaningful without the practice of gameplay' (2017: 6). By analysing the example of the *Dark Souls* series, they also focus on the indeterminate nature of the player-game relation, emphasizing the fact that these games' multiplayer features are, in fact, indeterminate – multiplayer mechanics may just manifest themselves while the player is focusing on their own single-player gameplay, and, in some situations, she might not even be aware that she is experiencing a multiplayer mechanic. This example highlights another important point in Barad's theory about objects and subject. Because agency is not attached to any particular entities, and is instead conceptualized as a dynamic force acting between entities (Barad 2007: 178), the strict division between the object and subject is no longer needed. Everything is in flux through the process of intra-action: we can be controlling our playable character one minute, but, in the next minute, they can just do something unexpected and we turn from being an active subject to a powerless observer (de Petris et al. 2017).

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MEANINGFUL PLAYFULNESS OF THE INTRA-ACTIONS

The transformative power of intra-action seems to be well attuned with the concept of the bio-object and video games, but there are few differences that need to be addressed. When we compare what we know about the concept of the bio-object and the idea of intra-action, it seems that the aforementioned rivalry inside the bio-object can be interpreted as this dynamic force of intra-action that is constantly changing in intensity, which results in the shifting of the boundaries between, and the changing properties of, the player and the game. However, it is happening on a different level of relation. The intra-action is connected strictly to matter and explains how it is in a constant state of becoming on an atomic level. A good example is the instance of the glitch, which could be interpreted as the manifestation of the game's agency (Janik 2017), but which, in a Baradian context, would instead be understood as the manifestation of the game's autonomy that appears as a result of the act of play with the player. The game's borders have been broadened – even if only for a moment, since these borders will be redrawn again in the moment when the player regains their dominant position by, for example, using the glitch to their advantage, or no longer being spooked by the aforementioned moving mannequins in *Skyrim*.

This aspect of the bio-object seems to be very well attuned to Barad's agential realism, but it continues this relation on a more functional and aesthetic level. Kantor and Barad appear to be in agreement on the fact that meaning is created through specific material-discursive practices (Barad 2007: 148; Kantor 2004: 396–98). However, what seems special about the idea of the bio-object, and what Barad does not examine at length in her work, is the different status of the object of aesthetic consideration. While her theories are very inspirational for artists and scholars interested in new materialist paradigms as well as multispecies and ecocritical aesthetics (see, as an example, dedicated issues of *Antennae*), art is not a main subject of her deliberation.

This does not mean that it is not possible to trace insights in Barad's work that have a bearing on aesthetic thought. In her essay about building a Baradian approach to multispecies aesthetic, Madeline Boyd (2015) takes inspiration from Barad's quote about representation:

Representation has confessed its shortcomings throughout history: unable to convey even the palest shadow of the Infinite, it has resigned itself to incompetence in dealing with the transcendent, cursing our finitude. But if we listen carefully, we can hear the whispered murmurings of infinity immanent in even the smallest details. Infinity is the ongoing material reconfiguring of nothingness; and finity is not its flattened and foreshortened projection on a cave wall, but an infinite richness.

(2012: 16–17)

On this basis, Boyd argues that the focal point of a posthuman aesthetics would be to open oneself to the possibilities of intra-action, and not to represent just a subjective point of view. At first glance, it seems difficult to align this with Kantorian theatre, especially considering the central, often autobiographical and active position that he put himself in during the performances, not to mention his controlling character (see Pieniążek 2005; Kobińska 2009). However, when we look closer into his artistic process, an intra-active openness can be spotted. Not only he was actively fighting ideas of representation

(and fiction) in his plays (Pleśniarowicz 1990; Kłossowicz 1991), but they were also pointedly open to their surrounding reality: he not only put a lot of attention into the objects that he implemented into his art, as I have mentioned before, but also responded to what reality gave him in an inclusive manner – for example, when one of his actors had to wear a neck brace, that neck brace become not just something to ignore, but an integral part of the bio-object (Welmiński 2014: 49).

However, what differentiates his concept of the bio-object from a potential Baradian aesthetics is the approach to the object, which Kantor termed 'L'OBJET D'ART'. While potentially every object can become a part of a bio-object, first, it needs to be subjected to a special aesthetical transformation. Therefore not every connection between a human and a non-human actant can, in Kantor's terms, be classified as a bio-object. A specific material-discursive practice is required in order for such a connection to result in the establishment of a bio-object. I would argue that, in the context of a video game, play would constitute such a practice. Using a Kantorian perspective, we can perceive the game object as something that is given, ready-made, but also something that is filled with possibilities waiting to be uncovered through the intra-action of play, but it exceeds the molecular transformation level that Barad focuses on. Like Kantorian objects and *emballages*, the game object was designed to convey aesthetical meanings. It is not art we are ready to prepare, but art we are playing with. Thanks to that, it is easier to see the game object in a more specific way that points our attention directly to the correlation between game and player inside their relation.

One can say that by combining Kantor's writing with the concept of intra-action, we are misreading Barad's theory to think more about poetics, about how aesthetic meaning can emerge through what she calls 'material-discursive practices'. However, by using the two conceptualities together, we are able to see the whole process of play: from the mattering of the matter, where both bodies are being transformed on a molecular level, to the intra-action of play interpreted in the context of a very specific form of theatre, which lets us continue the analysis of the relationship between player and the game on the symbolic level. This continuation is possible, because Kantor was interested in what the relation between human and non-human can tell us about very specific and hard-to-grasp ideas, like real(ness), illusion, memory, death and love. Those are important categories for us, as humans, but what Kantor's theatre sought to reveal was the extent to which they are always co-created to the same degree by objects.

Using those two terms together, we do not have to think about meanings only in the context of transformation in one or the other actant – about something that appears, in a way, 'inside' or within the borders of the actant. We can focus on meanings as emerging through the bio-object connection, because of the intra-actively created possibilities for that in the properties of human and non-human elements. A good example of this is fiction (understood as story) that, as Kantor (cited in Kobialka 2009) put it, is 'continuously disappearing and reappearing, "[shining] through" the "life of these bio-objects"' (Kobialka 2009: 359). The fiction, or for that matter any kind of narrative in a game, does not have to appear if there is no need, no response from both sides of the bio-object. Therefore, we can play *Skyrim* as an epic Dragonborn, role-playing every quest, but in another play-through we can just speedrun through the entire game, with the fiction not being a part of our experience.

A shift can also happen during the same play experience, when the action of the game imposes an undesigned event that becomes a part of a story in the player's eyes. As an example, I will use my own experience of playing *Skyrim*. During the first few hours of gameplay, I did not know all of the mechanics yet. My character suddenly fell ill, and I thought that a good night's sleep would cure her. After a long search, I finally found a bed inside a keep that my character could lay down in, only to realize in the morning that it was a double bed belonging to the Jarl of the hold and that, unknowingly, I had just let my character spend a night with him. An action that started as strictly a gameplay one, then, triggered an undesigned fictional event.

In a sense, any form of interpretation resulting from the playing of a game can be understood as an example of the meanings that can emerge from the tension inside the player–game bio-object. While most of the time interpretations retain the form of indeterminate phenomena (like thoughts or impressions), they can become more concrete because of the specific intra-actions of human and non-human that form the bio-object. But can the bio-object connection exceed the intra-action of play? At first glance, in both cases – Baradian concept of intra-action and Kantor's theatre – when the bond is severed, the transformative tension that generates the meanings should no longer be active. Anything that happens outside the bio-object does not belong to it. It is a separate intra-action. However, if we look closely into both theories, the connection lasts, but has different qualities. After stopping playing the game, the player does not lose their identity as a gamer. They might no longer be connected with the game object on the physical level, but feelings and thoughts, as well as skills, stay within them, waiting for another connection to happen. A similar phenomenon can be observed in Kantor's theatre. While the bio-object, as a connection between human and non-human actants, only lasts for the duration of its turn on the stage, the imprints of this bond endure. Kantor's objects continue to produce meanings, even in the form of copies or presented in a museum space, because of their special, 'poor' materiality that constantly performs itself in different contexts. Even without the presence of the human actors (Fazan 2019: 395–418), the object retains the trace of the former bond.

This situation of a constant reminiscence and repurposing of the old connections and situation is also visible on the human side of the equation. Other artists, inspired by Kantor's aesthetical actions, have tried to capture his absent presence in their new artistic performances and installations. It is not about recreating the old performances, but rather experimenting with past ideas, keeping them in a state of constant movement that evokes new meanings (Fazan 2019: 476–506).

The same goes for any kind of interpretation that originates from the bio-object bond during the intra-action of play. Our convictions, desires or experiences acquire a more determinate shape the moment they meet the materiality and processes of the game. The bio-object leaves an imprint both on the player and on the game object, allowing them to continue to perform meanings outside this bond. There is no shortage of explicit examples of this, including all kinds of fan works – fan arts, fan fictions, fan videos – or just simple comments on discussion threads and so on. Scanning through websites like reddit or, for that matter, any fanwiki, we can find a lot of discussions concerning certain aspects of a games that were unclear – for an example of this, we can consider the various findings or theories about Doctor Gaster from the role-playing game *Undertale* (Toby Fox 2015). He is a mysterious character

whose history needed to be uncovered by the collective work of dedicated players who were willing to delve deep into the game environment and the game files. In this way, the interpretation of the game expands from a simple walkthrough to a complex theory.

However, it does not have to be that elaborate. More straightforward examples could include all discussions regarding a particular character and players' feelings towards them, like all the comments about the love interests of Geralt in *The Witcher 3: Wild Hunt* (CD Projekt RED 2015) – especially Triss and Yennefer – in which players exchange their opinion about the motivations behind the choices they made regarding the witcher's love life, alongside detailed instructions on what to do – in terms of game mechanics – in order not to lose your chance with the characters.

Even if it does not seem like it, the video game object is always involved in the processes that lead to any interpretations. Not only because the intra-action of play triggers certain audio-visual messages designed in the game environment, but also because the game object actively adds something of its own. Once again, the glitch seems to be one of the phenomena that allows us to see this process of the creation of new meanings very clearly: the glitch is (in most cases) the outcome of the player–game intra-actions, and it is a manifestation of the game object's autonomy. While there is still a rivalry inside the bio-object, as I mentioned before, the player wants to regain the dominant position. One way of dealing with a glitch is the counterplay of glitching – the usage of glitches to gain advantage in the game – but we can also find a lot of glitch-related fan works, which seem to be an attempt to rebuild our position as more 'agential' – at least in our head. For example, because of the glitched software, people who played *Pokémon Go* (Niantic 2016) could approach a pocket monster that consisted of two different figures glitching into one. This situation triggered the imagination of some players to draw pictures of the quite horrifying resulting hybrids, which transforms the unprogrammed software behaviour into a worldbuilding experience.

This kind of interpretative meaning generation is by no means restricted to players' attempts to make sense of glitches. To find another category of non-glitch-based examples, we can turn to erotic fan works, like *Overwatch* (Blizzard Entertainment 2016) porn, that result from the encounter between the game's audio-visual and narrative properties and the player's desires. Of course, video games are not the only pop culture artefacts that generate this kind of content, but it is significant that no other non-game-related fictional characters made the list of top 20 search in 2018 on Pornhub. That might suggest at least two things: that video games have increased their popularity among the Pornhub audience (which is also indicated by Pornhub analytic data; see Pornhub Insights 2018a: n.pag.); and/or that the ludic properties of playable characters create the possibilities to be explored in the realms of adult content. The other interesting fact to note is that during the 24-hour outage of *Fortnite Battle Royale* (Epic Games 2017) servers in April 2018, searches for *Fortnite*-related pornographic content jumped by 60 per cent (Pornhub Insights 2018b: n.pag.), which might be interpreted as a search for a substitute for the meanings normally generated and satisfied by the game object.

Moreover, meaning and interpretation do not have to manifest only in fan culture, but also in other cultural domains – including academia. A good example is Olli Tapio Leino's engagement with the death loop he experienced while playing *Fallout: New Vegas* (Obsidian Entertainment 2010): his development of a theory of the game object as a playable artefact on the basis of this

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experience (Leino 2012) can, in this sense, be understood as an attempt to regain the dominant position in a situation where the game had left him with little to no control over his activity.

This specific situation of emerging meanings is also possible because of the nuanced ontological status of the bio-object. Even if the human and non-human actants appear as one, the bio-object is neither the object/game nor the actor/player. It emerges through the actions of both, and connects them on the level of their different materialities (respectively both digital and physical) through performative, meaning-generative processes (or intra-actions). In order for this to happen, human and non-human actants need to remain linked but distinct. In other words, while constituting the bio-object, the human being does not transcend their human condition, and the game object does not stop being a digital object. This also follows the premises of Joanna Zylińska's (2015) posthumanistic bioethics, in which she combines Levinas and Derrida to show how being-in-difference with non-humans would help us to better think about the Other we are facing in a situation of mutual connection and influence (Zylińska 2015: 211–30). Preserving one's uniqueness is important, because it allows us to establish relations with other beings and communicate with them.

In this line of thought, the game object becomes the Other, that is not only with us during the intra-action of play, but also makes us responsible for itself and our actions towards it. The notion of the bio-object helps us to see the duality of the game object as, on the one hand, a playable artefact (Leino 2012), in which digital materiality and processes are intertwined, and, on the other hand, a space for material-discursive practices where meanings are produced. The intra-action of play, which includes not only the audio-visual output and the actions that can be made in the environment, but also all the resulting sensations, both physical and mental, experienced by the player, becomes an act of communication (Majkowski 2019). I would argue that you can see this otherness in every game object; however, it is usually hidden, in order to build a cohesive illusion of the game as an environment or world. It is easier to spot in games like *Undertale* or *Doki Doki Literature Club* (Team Salvato 2017), where the diegetic layer of the game is intertwined with the non-diegetic layer. In both cases, some characters in the game – like Flowey in the former or Monika in the latter – are aware of what they are, and their position is therefore different from that of other characters. They can, for example, manipulate the game files or gather data about the player. In both cases, the player has to face not so much a single persona, but rather the game object that hides under that character's mask. In the final moments of both games, the game object/character blocks the player from either loading their previous saves or even finishing the game if they do not delete certain files. We are not just playing around IN a game environment, we are playing around WITH the game object. The game becomes our partner in play – the Other imposes its otherness over us.

CONCLUSION

Using the bio-object idea in relation to the paradigm of critical posthumanism gives us an opportunity to 'hear' the voice of the game object, which can sometimes be hidden beneath its anthropocentric design. Analysing the video game within this framework helps us understand how the game object and the player not only influence each other, but become partners in creating

meanings. This approach can also lead to new perspectives on some of the key concepts of game studies, especially agency, which, viewed from Barad's perspective, should be redefined in the context of video games. As it is not something that an entity possesses, there is no need for attributing agency to one entity or another, or considering when agency can be manifested. Instead, agency is always present, because without its intra-active power, the game and/or the player would simply not be the game and the player. There are already some works that consider agency in relation to games from the perspective of posthumanism, like Daniel Muriel and Gary Crawford (2020), who try to explore this concept using Latourian assemblages. However, most existing research about agency in games is, in most cases, strictly focused on players and their needs. Accordingly, it is either perceived as the player's ability and pleasure that can be generated during the gameplay experience (Murray 1997), a possible feature that can be experienced by the player through specific affordances (Mateas 2004; Wardrip-Fruin et al. 2009), or a problem-solving activity and goal-oriented process (Jørgensen 2003). Even if there is research acknowledging that the game or the objects inside the game environment can also be considered to have a form of agency, it is usually understood as different, if not incomplete, compared to human agency. For example, for Susanne Eichner, the main trait of the player's agency is meaning-making, which is reserved, for now, only for humans (2014: 41). As I have hopefully shown in this text, meanings are something that emerge from the intra-actions between the player and the game, and we cannot pinpoint one definite source of them. By following Barad's idea of agency, we escape a sometimes involuntarily anthropocentric approach where we either do not grant non-human actants the rights to have agency, or create different categories that exclude some actants from having the same level of influence on reality. The combination of the idea of the bio-object and the intra-active power that bonds it is what makes the player and the game equally important co-creators of meaning, on every level and at every stage of the connection – from the depths of the materiality of the respective bodies, to the interpretational spaces outside the bio-object connection.

The notion of the bio-object, then, not only allows us to see concepts like agency and meaning-making in a new, posthuman light, but also emphasizes the fact that, despite the intra-active practice of play bonding them together, the game and the player still remain separate entities with separate properties. The ontological difference remains valid, rather than, as in the case of some approaches in cyborg theories, either becoming irrelevant or continuing to concentrate on a human centre (Zylinska 2015: 212). Without this difference, there would not be any tension that would generate meaning. Therefore, by using Kantor's idea of the bio-object and Barad's concept of intra-actions, we are not only creating analytical tools to better understand the relation between the player and the game object, but also shifting our perception of play, coming to understand it as being always an aesthetic as well as ethical experience of communication with the Other.

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